Yao **Teng**

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Pokfulam, Hong Kong

Education

Sept. 2023 - Present	**PhD in Electrical and Electronic Engineering**, The University of Hong Kong Supervisor : Prof. Xihui Liu	Hong Kong, China
Sept. 2020 - Jun. 2023	**M.Sc. in Computer Science and Technology**, Nanjing University M.Sc. in MCG Lab, Supervisor : Prof. Limin Wang	Nanjing, China
Sept. 2016 - July. 2020	**B.S. in Software Engineering**, Xidian University **GPA** : 3.7/4.0 | **Rank** : 18/445 (4%)	Shaanxi, China

Research Interests

Computer Vision : image & video generation, visual detection, scene understanding

Projects & Internships

Feb. 2026 - Present **Tencent Hunyuan World Model Team** Core Contributor of Hy-Nomas

- **VAE training framework.** Built the team's image/video VAE training codebase, supporting (i) distributed training that shards both model parameters and activations across GPUs, enabling multi-scale, image-video joint training on high-resolution long videos; (ii) mainstream VAE architectures with standard training objectives, including perceptual, GAN and semantic losses; (iii) standardized reconstruction benchmarks and training-health monitoring; and (iv) billion-parameter DiT pre-training for latent-space validation.
- **Billion-scale data curation.** Built an image/video pre-training data curation pipeline on a hundred-GPU cluster, covering large-scale embedding extraction, hierarchical clustering, deduplication, category rebalancing and safety filtering.
- **One-step pixel diffusion decoder.** Building on the framework and data above, trained a ∼7B pixel-space video DiT as the VAE decoder, compensating for high-frequency details lost in latent compression; distilled it into a one-step sampler and enabled streaming decoding for world-model generation. *Integrated into the final mainline release.*

Mar. 2025 - Jan. 2026 **ByteDance TikTok** Research on Video VAE and Text-to-Video Generation

Publications

- **Yao Teng**†, Guoxuan Chen†, Yimeng Chen*, Han Shi*, Piotr Piękos, Jie Ran, Yutao Zeng, Chuanyang Zheng, Jingyao Li, Li Nanbo, Haoli Bai, Chao Huang, Xihui Liu, Di He, Weiyang Liu and Jürgen Schmidhuber. **Towards Next-Generation Architectures of Large Language Models : A Survey**, Under Review, 2026.
- **Yao Teng**, Minxuan Lin, Xian Liu, Shuai Wang, Xiao Yang, Xihui Liu. **Adaptive 1D Video Diffusion Autoencoder**, Under Review, 2026.
- **Yao Teng**, Zhihuan Jiang, Han Shi, Xian Liu, Xuefei Ning, Guohao Dai, Yu Wang, Zhenguo Li, Xihui Liu. **SJD++ : Improved Speculative Jacobi Decoding for Training-free Acceleration of Discrete Auto-regressive Text-to-Image Generation**, IEEE Transactions on Pattern Analysis and Machine Intelligence (**T-PAMI**), 2026.
- **Yao Teng**, Fuyun Wang, Xian Liu, Zhekai Chen, Han Shi, Yu Wang, Zhenguo Li, Weiyang Liu, Difan Zou, Xihui Liu. **Speculative Jacobi-Denoising Decoding for Accelerating Autoregressive Text-to-image Generation**, in The Thirty-Ninth Annual Conference on Neural Information Processing Systems (**Neurips**), 2025.
- **Yao Teng**, Han Shi, Xian Liu, Xuefei Ning, Guohao Dai, Yu Wang, Zhenguo Li, Xihui Liu. **Accelerating auto-regressive Text-to-Image Generation with Training-free Speculative Jacobi Decoding**, in The Thirteenth International Conference on Learning Representations (**ICLR**), 2025.
- Shuai Wang, **Yao Teng**, Limin Wang. **Deep Equilibrium Object Detection and Segmentation**, IEEE Transactions on Pattern Analysis and Machine Intelligence (**T-PAMI**), 2025.
- Yuqing Wang, Zhijie Lin, **Yao Teng**, Yuanzhi Zhu, Shuhuai Ren, Jiashi Feng, Xihui Liu. **Bridging continuous and discrete tokens for autoregressive visual generation**, in IEEE International Conference on Computer Vision (**ICCV**), 2025.
- Kaiyue Sun, Xian Liu, **Yao Teng**, Xihui Liu. **Personalized text-to-image generation with auto-regressive models**, Under review, 2025.
- Fu-Yun Wang, Keqiang Sun, **Yao Teng**, Xihui Liu, Jiaming Song, Hongsheng Li. **Self-NPO : Negative Preference Optimization of Diffusion Models by Simply Learning from Itself without Explicit Preference Annotations**, Under review, 2025.
- Wanghan Xu, Xiaoyu Yue, Zidong Wang, **Yao Teng**, Wenlong Zhang, Xihui Liu, Luping Zhou, Wanli Ouyang, Lei Bai. **Exploring representation-aligned latent space for better generation**, Under review, 2025.
- **Yao Teng**, Yue Wu, Han Shi, Xuefei Ning, Guohao Dai, Yu Wang, Zhenguo Li, Xihui Liu. **DiM : Diffusion Mamba for Efficient High-Resolution Image Synthesis**, Under review, 2024.

- **Yao Teng**, Enze Xie, Yue Wu, Haoyu Han, Zhenguo Li, Xihui Liu. **Drag-a-video : Non-rigid Video Editing with Point-based Interaction**, Under review, 2024.
- Yisen Wang, **Yao Teng**, Limin Wang. **CycleHOI : Improving Human-Object Interaction Detection with Cycle Consistency of Detection and Generation**. Under review.
- Liang Zhao*, **Yao Teng***, Limin Wang. **Logit Normalization for Long-tail Object Detection**. **IJCV**.
- Haisong Liu, **Yao Teng**, Tao Lu, Haiguang Wang, Limin Wang. **SparseBEV : High-Performance Sparse 3D Object Detection from Multi-Camera Videos**, in IEEE International Conference on Computer Vision (**ICCV**), 2023.
- Shuai Wang, **Yao Teng**, Limin Wang. **Deep Equilibrium Object Detection**, in IEEE International Conference on Computer Vision (**ICCV**), 2023.
- **Yao Teng**, Haisong Liu, Sheng Guo, Limin Wang. **StageInteractor : Query-based Object Detector with Cross-stage Interaction**, in IEEE International Conference on Computer Vision (**ICCV**), 2023.
- **Yao Teng**, Limin Wang. **Structured Sparse R-CNN for Direct Scene Graph Generation**, in IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2022.
- **Yao Teng**, Limin Wang, Zhifeng Li, Gangshan Wu. **Target Adaptive Context Aggregation for Video Scene Graph Generation**, in IEEE International Conference on Computer Vision (**ICCV**), 2021.

Skills

Programming Skills : Python, LaTeX, C/C++, Matlab, Java.
Languages : Mandarin, English.

Honors & Awards

2021	**National Scholarship for Postgraduates**	Nanjing University
2021	**Pacemaker to Outstanding Graduate Student**	Nanjing University
2021	**1st Award**, Scholarship for Postgraduate Students	Nanjing University
2019	**Silver Medal**, The 2019 ICPC China Shaanxi Provincial Programming Contest	Xidian University
2018	**Bronze Medal**, The 2018 ACM-ICPC China Invitational Contest Xi'an Site	Xidian University